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Studierichting: Informatica
Oefeningenreeks 4A nr. 46.3

$$\int \frac{3x^2 + 6}{x^4 + 5x^2 + 4} dx = \int \frac{3x^2 + 6}{(x^2 + 1)(x^2 + 4)} dx$$

we splitsen eerst de partieelbreuken:

$$\frac{3x^2 + 6}{(x^2 + 1)(x^2 + 4)} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{x^2 + 4}$$

$$A(i) + B = \left. \frac{3x^2 + 6}{x^2 + 4} \right|_{x=i} = 1 \text{ dit impliceert dat: } A = 0 \text{ en } B = 1$$

$$C(2i) + D = \left. \frac{3x^2 + 6}{x^2 + 1} \right|_{x=2i} = 2 \text{ dit impliceert dat: } C = 0 \text{ en } D = 2$$

$$\begin{aligned} I &= \int \frac{1}{x^2 + 1} dx + 2 \int \frac{1}{x^2 + 4} dx \\ &= \text{Bgtan}(x) + \text{Bgtan}\left(\frac{x}{2}\right) + c \end{aligned}$$

References